	Asset A r_A %	r_B %	r_p E_p portfolio
scenario ~~option~~ 1	15	6	10.5
Option 2	9	6	7.5
Option 3	3	12.	7.5
	9	8	~~8.5~~ 8.5

$i = A$ or B

$m = 1, 2, 3$

$\frac{25.5}{3} = 8.5$

$$\boxed{E[r_i] = \sum_{m=1}^{3} P(\text{scenario}_m) \cdot r_i(\text{scenario}_m)}$$

$$\Rightarrow E[r_A] = \tfrac{1}{3} \cdot 15 + \tfrac{1}{3} \cdot 9 + \tfrac{1}{3} \cdot 3 = \frac{27}{3} = 9$$

$$E[r_B] = \tfrac{1}{3} \cdot 6 + \tfrac{1}{3} \cdot 6 + \tfrac{1}{3} \cdot 12 = \frac{24}{3} = 8$$

$$\boxed{\sigma^2[r_i] = \sum_{m=1}^{3} P(\text{scenario}_m) \left[r_i(\text{option}_m) - E[r_i] \right]^2}$$

$$\sigma^2[r_A] = \tfrac{1}{3} \cdot [15 - 9]^2 + \tfrac{1}{3}[9 - 9]^2 + \tfrac{1}{3}[3 - 9]^2$$

$$= \tfrac{1}{3} 36 + \tfrac{1}{3} \cdot 0 + \tfrac{1}{3} 36 = \frac{2 \cdot 36}{3} = 24\%^2$$ (in reality 0.0024)

$$\sigma^2[r_B] = \tfrac{1}{3}[6 - 8]^2 + \tfrac{1}{3}[6 - 8]^2 + \tfrac{1}{3}[12 - 8]^2$$

$$= \tfrac{1}{3} \cdot 4 + \tfrac{1}{3} \cdot 4 + \tfrac{1}{3} \cdot 16$$

$$= \frac{24}{3} = 8\%^2$$ (0.0008)

$$\boxed{E[\text{portfolio}] = E\left[\tfrac{1}{2} r_A + \tfrac{1}{2} r_B \right] = \tfrac{1}{2}\left(E[r_A] + E[r_B] \right)}$$

f_A, f_B

$= 8.5$

$$\boxed{\text{s.t. } f_A + f_B = 1}$$

better below than above

$$\boxed{E[\text{portfolio}] = \sum_{m=1}^{3} P(\text{scenario}_m) [r_p(\text{scenario}_m)]}$$

$$\sigma^2[\text{portfolio}] = \sum_{m=1}^{3} P(\text{scenarios}_m)\left[r_p(\text{scenario}) - E[\text{portfolio}]\right]^2$$